

Atomic Structure & the Periodic Table

Quick reference sheet · Chemistry · keep handy for revision

1. Subatomic particles

Particle	Relative mass	Relative charge	Location
Proton	1	+1	Nucleus
Neutron	1	0	Nucleus
Electron	very small	-1	Shells around nucleus

Picture it: shrink the atom to the size of a football stadium and the nucleus is a pea on the centre spot, holding nearly all the weight. Protons (+) and electrons (-) wear opposite badges and attract; neutrons wear no badge (0) and just add weight.

2. Atomic number, mass number & isotopes

Atomic number = number of protons = number of electrons. **Mass number** = protons + neutrons. So neutrons = mass number – atomic number.

Isotopes: same element (same protons), different number of neutrons. Same chemical properties, different physical properties (mass, density; some radioactive).

Relative atomic mass = weighted mean of isotope masses, using % abundance: $A_r = \Sigma(\text{mass} \times \% \text{abundance}) \div 100$.

3. Electronic structure

Electrons fill the lowest-energy shell first. Maximum per shell (first 20 elements): **1st = 2, 2nd = 8, 3rd = 8**. Example: sodium (11) = 2,8,1.

Group number = electrons in outer shell. **Period number** = number of occupied shells.

4. Development of the periodic table

Newlands (1864): ordered by atomic weight; "Law of Octaves" — every 8th element had similar properties. Rejected: forced dissimilar elements together, no gaps for undiscovered elements.

Mendeleev (1869): ordered mostly by atomic weight but swapped some elements to group similar properties together; left gaps and predicted missing elements' properties — later discoveries (e.g. germanium) matched, giving the table credibility.

Modern table: ordered by increasing atomic (proton) number. Same group = same outer electrons = similar properties. Same period = same number of shells.

5. Metals vs non-metals

	Metals	Non-metals
Position	Left & bottom-left	Top-right
Properties	Shiny, malleable, good conductors, high m.p./b.p.	Dull, brittle, poor conductors, low m.p./b.p.

Ion formed	Positive (lose electrons)	Negative (gain electrons) or covalent
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6. Groups 0, 1 & 7

Group	Outer electrons	Trend	Key facts
0 — Noble gases	Full shell	B.p. increases down group	Unreactive, monatomic gases
1 — Alkali metals	1	Reactivity increases down group	Soft, low density, react vigorously with water
7 — Halogens	7	Reactivity decreases down group	Diatomic molecules, can displace less reactive halogens

Trend reason: going down a group, the outer shell is further from the nucleus and more shielded — easier to lose an electron (Group 1, reactivity up), harder to gain one (Group 7, reactivity down).